Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/09/2018
PI Name:	Asaithamby, Aroumougame Ph.D.		
Project Title:	Mechanism of Clustered DNA Double-strand Break Repair in Response to HZE Particles in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Dallas	**State:**	TX
Zip Code:	75390-7201	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2017 per S. Monk/LaRC (Ed., 11/7/16) NOTE: Extended to 12/31/2016 per NSSC information (Ed., 10/5/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chen, David Ph.D. (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX13AD57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	DNA damage and consequent mutations initiate the multi-step carcinogenic process. Substantial evidence indicates that high charge and energy [HZE] particles, component of galactic cosmic rays (GCR), induced clustered DNA damages are difficult to repair, and in some instances are irreparable; this has been associated with the increased relative biological effectiveness of chromosomal aberrations, mutagenesis, and carcinogenesis. However, it is unclear why clustered lesions are difficult to repair, and the mechanisms by which unrepaired clustered DNA lesions trigger carcinogenic events are not known. The goals of this proposed research are to utilize our unique capability to quantitatively estimate the ability of cells to repair clustered DNA lesions in response to a spectrum of HZE particles, and to dissect the genetic factors that may potentially be involved in the processing of complex DNA lesions. The specific aims of the proposed research are to test the hypothesis that: (1) the cellular ability to repair HZE particles induced clustered DNA damage is modulated by radiation quality and cell cycle stages, utilizing our newly established in situ and imaging technology; (2) homologous recombination plays a key function in facilitating the repair of double-strand breaks associated with clustered DNA lesions, using key homologous recombination factors deficient cells; (3) Fanconi Anemia (FA) complex may orchestrate the repair of complex double strand breaks (DSBs) induced by HZE particles, using a panel of FA pathway deficient human cells; and (4) DNA end-processing nucleases, such as Werner Syndrome protein (WRN) and Artemis, may facilitate the processing of complex DNA ends generated by HZE particles using human cells that are either deficient in WRN and Artemis or expressing different nuclease mutant forms of WRN and Artemis. Understanding the mechanism underlying the repair of DNA damage induced by HZE particles will have important implications for estimating the risks to Astronaut health associated with GCR exposures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the mechanism underlying the repair of DNA lesions induced by HZE particles will have important implications for estimating the risks to Astronaut's health associated with GCR exposures. Our research would also lead to the development of counter-measures to enhance astronaut’s ability to sustain HZE particles exposures in space.
Task Progress:	During the final year of this grant, we identified a novel mechanism by which space radiation triggers immune signaling in human cells. We conclude that the immune signaling initiated by the space radiation will contribute to the carcinogenesis risks. 1. RAD51 is critical for complex DSB (double strand break) repair in S/G2 cells in response to Fe-particle irradiation. 2. RAD51 blocks MRE11-mediated degradation of nascent DNA strands in response to Fe-particle exposure. 3. RAD51 depletion upregulates expression of innate immune signaling genes by accumulating self-DNA in the cytoplasm of S/G2 phase cells. 4. RAD51 suppresses chromosome breakages following Fe-particle exposure.
Bibliography Type:	Description: (Last Updated: 03/12/2018)
Articles in Peer-reviewed Journals	Bhattacharya S, Srinivasan K, Abdisalaam S, Su F, Raj P, Dozmorov I, Mishra R, Wakeland EK, Ghose S, Mukherjee S, Asaithamby A. "RAD51 interconnects between DNA replication, DNA repair and immunity." Nucleic Acids Res. 2017 May 5;45(8):4590-605. https://doi.org/10.1093/nar/gkx126 ; PubMed PMID: 28334891; PubMed Central PMCID: PMC5416901 , May-2017
Articles in Peer-reviewed Journals	Bhattacharya S, Asaithamby A. "Repurposing DNA repair factors to eradicate tumor cells upon radiotherapy." Transl Can Res. 2017 Jul;6 Suppl 5:S822-S839. https://doi.org/10.21037/tcr.2017.05.22 , Jul-2017